

Instantaneous stress detection from physiological and physical markers in shooting simulation

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Introduction: Detecting or predicting stress in day-to-day scenario is becoming attainable and ubiquitous due to the enormous data which can be extracted from smartphones [1] and other low-cost sensors [2]. But detecting stress in flight or fight situations or in very small duration of time is still an open research topic because of the detecting complications of stress.

Problem statement: Instantaneous stress detection is a process of stress prediction using data of a very small period of time Δt (typically $\Delta t < 10$ seconds). In Eclipse lab, we collected data of various sensors (Accelerometer, Heart rate, Blood pressure, Galvanic skin response) during a shooting simulation game for 10 subjects. The game session was consisted of 6 sessions each with a low stress and high stress condition. In low stress condition, the shooting target appears slowly and for high stress condition, it appears faster. In this project, we sought to predict the stress of the subject in between the time when the target is visible in the simulation screen.

Prior art: Similar work of prediction of stress was performed by Dobbins at el [3] but they implemented the whole system for driving scenarios. Moreover, Castaldo at el [4] claims that for ultra short period, HRV (Heart rate variability) is a significant physiological marker for stress.

Problem formulation: For detecting instantaneous stress, we need to filter and remove outliers from raw ECG (Electrocardiogram) data, and then we need to calculate the heart rate variability of the subject within the session as per Figure 1

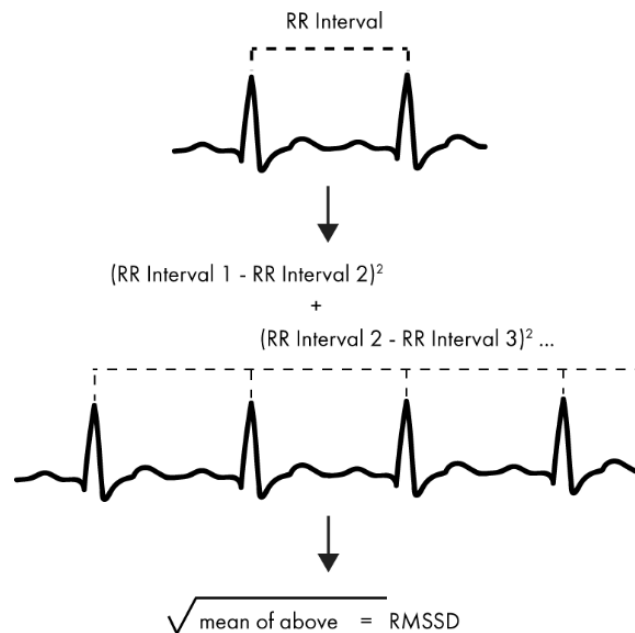


Figure 1

Then, by extracting the RMSSD (Root Mean Square of the Successive Differences) in between the timestamps of the target screen time we can predict the state of the subject in that exact situation.

We also plan to process the other sensors data like Accelerometer, GSR as well.

Plan of work: The plan of work is as follows

1. Preprocess the ECG, GSR, Accelerometer data (Filtering and Outlier removal)
2. Extract RMSSD from ECG data
3. Extract other sensors reading (processed) in the exact timestamp when a target is present on-screen.
4. Normalize the data and apply machine learning algorithms.

References:

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